



RESEARCH PROPOSAL

IT41043 — Intelligent Systems

***A Lightweight CNN-Based Food Freshness Detection System Using
Smartphone Images for Sri Lankan Market Environments***

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SECTION A — Research Gap Statement

With rising concerns of food safety, food waste and food quality control, the field of food quality assessment is an important research area in artificial intelligence and computer vision. The traditional food inspection techniques are predominant based on human visual inspection, which is often subjective and inconsistent. With the advancements of deep learning and image processing, automated freshness detection systems have become possible, especially for image classification tasks with Convolutional Neural Networks (CNNs).

CNN-based models like ResNet, MobileNet, VGGNet and EfficientNet have been previously used for freshness classification of fruits and vegetables. These studies reached high accuracy with the help of transfer learning, data augmenting and controlled datasets. But, most of the current systems are trained by taking the image under laboratory conditions where the light is stable and the background is clean and not in real market conditions.

The evaluation of freshness in Sri Lankan local markets is largely manual, and small vendors do not have the resources to purchase high-priced imaging systems. Smartphone based solutions are more functional and readily available. So this research suggests a light-weight CNN based food freshness detection system based on smartphone image acquired in real market condition in Sri Lanka. The study will assess the performance of the lightweight models of transfer learning against a traditional CNN baseline in realistic environments.

SECTION B — Research Question

Research Question

Can a lightweight transfer learning based CNN model outperform a traditional CNN baseline model in the classification of food freshness using the smartphone images taken under real Sri Lankan market conditions?

Sub-Questions

1. What is the highest classification accuracy obtained for the detection of food freshness using the light weight CNN architecture?
2. What is the impact of the lighting variation and background noise in the real world on the model?
3. Is the performance of the food freshness classification model better when implementing transfer learning, than when using of a traditional CNN model?

- 4) Which is the most useful measure of the performance of the models proposed (accuracy, precision, recall or F1-score)?
- 5) Are the images collected using a smart phone sufficient for an accurate food freshness classification in local markets?

SECTION C — Scope Definition

A deep learning system based on the image information of a cell phone. In the study, the customised dataset of fruits and vegetables will be collected based on real market condition in Sri Lanka through the smartphone camera. There will be various freshness categories in the dataset: fresh, medium, rotten. The research process will involve image pre-processing, data augmentation, model training, and performance evaluation.

The study will assess how well a traditional CNN performs against the lightweight transfer learning models like MobileNetV2 and EfficientNetB0. Based on the classification results, standard classification metrics such as accuracy, precision, recall, F1-score and confusion matrices will be used for model evaluation.

This research excludes real-time deployment, mobile application development, cloud integration, multilingual interfaces and large-scale commercialization. The study only covers offline model development and performance comparison.

SECTION D — Annotated Bibliography

[1] Howard et al., “MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications”, 2017.

Summary:

The authors introduce a new lightweight CNN model for mobile and embedded vision applications, called MobileNet, that is based on depth-wise separable convolutions in this paper.

Relevance:

The study is relevant because the developed MobileNet is an efficient deep learning architecture, which can be used for freshness detection systems on mobile phones.

[2] Tan and Le, The efficientnet: Rethinking model scaling for convolutional neural networks, 2019.

Summary:

The authors introduce EfficientNet, a CNN structure that enhances the model performance by compound scaling.

Relevance:

This paper is relevant since EfficientNetB0 will be used as a lightweight transfer learning model in the proposed research.

[3] Krizhevsky et al., “ImageNet Classification with Deep Convolutional Neural Networks”, 2012.

Summary:

The findings of this research indicate that deep CNNs can be used for large-scale image classification tasks.

Relevance:

The paper introduces the basic concepts of CNN image classification, which is used in this research.

He et al., 2016, Deep Residual Learning for Image Recognition.

Summary:

The paper presents ResNet architectures tackling the degradation issue in the deep NN.

Relevance:

This study is a part of the research to understand the transfer learning techniques in image based classification systems.

Too et al., 2019, A comparative study on the fine-tuning of deep learning models for identification of plant diseases.

Summary:

This study evaluates and contrasts different TL models for the classification of agriculture images.

Relevance:

The study offers comparative evaluation methods which are like the proposed freshness detection research.

[6] Hasan et al., “Deep Learning Approaches for Fruit Freshness Classification,” 2021.

Summary:

In this paper, CNN models for fruit freshness image classification with image datasets will be explored.

Relevance:

The research is directly related to food freshness detection and has useful experimental approaches.'

Rahman et al., 2020, "Fruit Quality Detection Using Convolutional Neural Networks".

Summary:

The authors propose a digital image based fruit quality identification system designed based on CNN.

Relevance:

The CNN architecture was supported in this paper for the food quality assessment by image.

[8] Shorten and Khoshgoftaar, “A Survey on Image Data Augmentation for Deep Learning” 2019.

Summary:

This survey covers different techniques for image augmentation for enhancing deep learning model performance.

Relevance:

The methods mentioned are applicable to improve classification with few data sets.

[9] Chollet, Xception: Deep Learning with Depthwise Separable Convolutions, 2017.

Summary:

The paper proposes a new architecture named Xception, which is based on the depthwise separable convolution operations.

Relevance:

This research helps the light-weight CNN concept for mobile-based applications.

[10] Very Deep Convolutional Networks for Large-Scale Image Recognition”, Simonyan and Zisserman, 2015.

Summary:

This study proposes the VGG architecture of the deep CNN layers for image recognition applications.

Relevance:

The paper gives background information about CNN structures in the image classification research.

SECTION E — Target Journal Justification

The proposed research is appropriate for Applied Sciences, as it is a research journal focused on artificial intelligence, machine learning, computer vision and intelligent systems. This study emphasizes the detection of food freshness through image analysis utilizing compact deep learning architectures, fitting with the journal's scope on applied AI and engineering applications. The journal is also suitable due to its focus on practical and real life technological solutions. The proposed research is to develop a low cost and accessible framework based on the images captured by smartphone under the realistic Sri Lankan market conditions to detect the freshness of food. This applied work is in line with the interest of the journal in applied intelligent systems and image analysis technologies. In addition, the journal offers open access publication options, that could improve the visibility and accessibility of the results of the research. Due to its moderate impact factor and interdisciplinary nature, it is appropriate for undergraduate level and early stage research in machine learning and computer vision.